

DS3040 Introduction to Deep Learning Lab

Assignment-6, Date: February 17, 2025

Timing: 10:00 to 12:45 PM

Max Marks: 20

Instructions

1. Submit one .ipynb file containing all answers. The name should be [student_name]_assignment[number].ipynb
 2. Write the questions in separate text blocks before the answers.
 3. Outputs for all sub-questions should be given and the code should be executable.
 4. Write justifications for your choices where needed.
 5. Ensure that all plots include clear labels and legends for better interpretation.
 6. Do not use AI or any websites except library documentations.
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1. Text Classification(10 marks)

Perform text classification using different recurrent neural network architectures.

- (a) Load the given data and do the necessary Preprocessing for text(including lowercasing, removal of punctuation and stopwords etc..) and labels.
- (b) Create a vocabulary based on the frequency of each word in the dataset, Convert the preprocessed text data into sequences of integers, where each integer corresponds to a word's index in the vocabulary.

Hint: *from tensorflow.keras.preprocessing.text import Tokenizer*

- (c) Apply padding to these sequences to ensure they have a uniform length.

Hint: *from tensorflow.keras.preprocessing.sequence import pad_sequences*

- (d) Develop and train three different types of neural network models for text classification: Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU).
- (e) Describe the effectiveness of each model (RNN, LSTM, and GRU) in performing the text classification task, comparing their performance.
- (f) Experiment with different hyperparameter settings on the best-performing model (RNN, LSTM, or GRU) by trying at least two different combinations, varying parameters such as the number of layers, embedding size, etc.

2. Next Word Prediction (10 marks)

Develop a next-word predictor using different recurrent neural network architectures.

- (a) Load the *shakespeare-hamlet.txt* from the *Gutenberg corpus* of NLTK and perform necessary preprocessing.

- (b) Tokenize the text, create word sequences, apply padding, and generate input sequence-next word pairs for training.
- (c) Construct and train a language model using RNN, LSTM, and GRU architectures.
- (d) Describe the effectiveness of each model (RNN, LSTM, and GRU) in performing the next word prediction task, comparing their performance.
- (e) Experiment with different hyperparameter settings on the best-performing model (RNN, LSTM, or GRU) by trying at least two different combinations, varying parameters such as the number of layers, embedding size, etc.